Lpts 317. 9/7.

Exaple of RegExp:

All strings that do not contain 111 as substrings.

Exaple string: 01100010

00...0 11 000...0 1 000...0

$$0^*((1+11)0^+)^*(\Lambda+1+11)0^*$$

encode a natural number into unsigned binary.

I want a regExp for all natural numbers divisible by 4.

Exple: $8 = 1000_2$

$12 = 1100_2$

RegExp: $1(1+0)^*00 + 0$

Show: $(L_1+L_2)^* = (L_1^*+L_2^*)^*$.

Proof. (1). LHS $\subseteq$ RHS. why? since

$$L_1 \subseteq L_1^*, \quad L_2 \subseteq L_2^*.$$

(2). Need show RHS $\subseteq$ LHS. why?

$$(L_1^*+L_2^*)^*$$

$$= \bigcup_{k\geq 0} (L_1^*+L_2^*)^k$$

Proof-writing.

Show $L^* L^* = L^*$.

Proof. $L^* L^*$

$$= \left(\bigcup_{k \geq 0} L^k \right) \cdot \left(\bigcup_{m \geq 0} L^m \right)$$

$$= \bigcup_{\substack{k \geq 0 \\ m \geq 0}} L^k \cdot L^m$$ // I take $n = k+m$

$$= \bigcup_{k, m \geq 0} L^{k+m} = \bigcup_{n \geq 0} L^n = L^*.$$ ∎

$$= \bigcup_{\substack{k \geq 0 \\ i_1, \dots, i_k \in \{1,2\}}} L_{i_1}^* \cdot L_{i_2}^* \cdots L_{i_k}^*$$

// Each of
// $L_{i_1}, \dots, L_{i_k}$
// $\subseteq L_1 + L_2$

$$\subseteq \bigcup_{\substack{k \geq 0 \\ i_1 \cdots i_k \in \{1,2\}}} \underbrace{(L_1+L_2)^* \cdots (L_1+L_2)^*}_{k \text{ times.}}$$

$$= \bigcup_{k \geq 0} \underbrace{(L_1+L_2)^* \cdots (L_1+L_2)^*}_{k \text{ times.}}$$

$$= \bigcup_{k \geq 0} (L_1 + L_2)^*$$

$$= (L_1 + L_2)^* = LHS. \qquad \blacksquare$$

structural Induction ——— a way to prove That a language is regular.

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math induction. (math 216).

How to prove:

$$\forall n \geq 0. \quad P(n)$$

Predicate on n.

(Example: $P(n) \equiv$ the sun rises from east at day n).

We can <u>only</u> do this:

(1). $P(0)$ holds;

(2). ~~For each $n \geq 0$,~~

if $P(n)$ holds, then $P(n+1)$ holds.

Induction Hypothesis.

$\Rightarrow \forall n.\ P(n)$.

Variations:

I.H. Assume that for all past days i (from today n), $P(i)$ holds ($\forall i < n$)

$\Rightarrow$ your job is to prove $P(n)$ holds for today n.